

6. [Maximum mark: 6]

The line L_1 has vector equation $\mathbf{r} = 4\mathbf{i} - \mathbf{k} + \lambda(a\mathbf{j} + \mathbf{k})$, where $a, \lambda \in \mathbb{R}$.

The line L_2 has vector equation $\mathbf{r} = \mathbf{i} - b\mathbf{k} + \mu(\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$, where $b, \mu \in \mathbb{R}$.

The lines L_1 and L_2 are perpendicular and intersect at a unique point.

Find the value of a and the value of b .

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